

Micro Hack 2 — SDC Workloads

Participant Workbook · GreenGrid Scorecard

FY27 EMEA EPS · Fabric Apps Motion

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Micro Hack 2 — SDC Workloads Workbook (Participants)

Track: **SDC Workloads (Fabric Extensibility Toolkit)** Scenario: **GreenGrid Analytics** (fictional SDC)

Print-ready PDF: [microhack-2-sdc-workloads-workbook.pdf](#)

This is a **didactic** micro-hack. You are **not** expected to *find* the solution — every step tells you **exactly what to type or paste, why, and what you should see**. Follow them in order and you get a working Fabric workload, starting from a clean machine.

Day agenda (1 day)

Morning — presentations & demos. Afternoon — hands-on hack. This is a **standalone one-day micro-hack** for the SDC Workloads (Extensibility Toolkit) track.

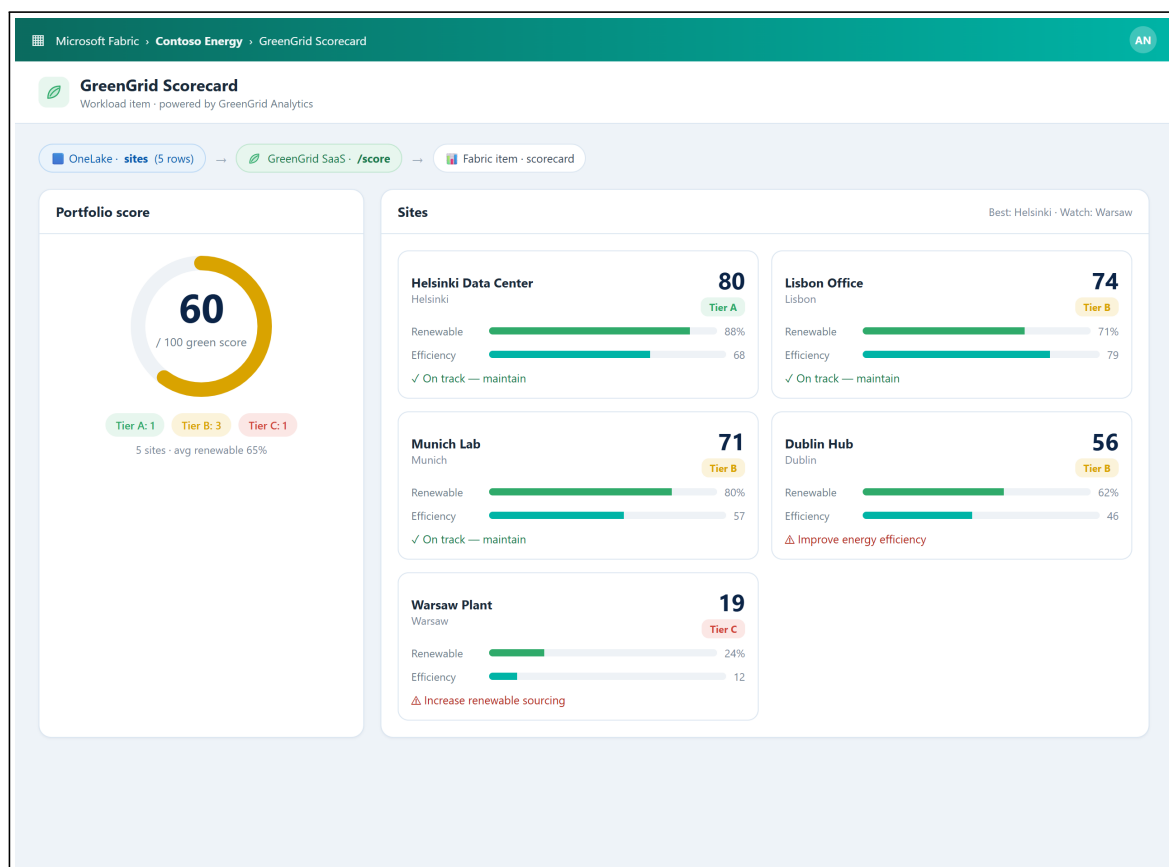
Time	Session
09:00	Welcome & motion overview — the Fabric apps motion for EMEA
09:30	Microsoft Fabric foundations — OneLake, capacity, workspaces
10:00	Extensibility Toolkit foundations — workloads, items, Dev Gateway
10:30	Break
10:45	Demo · GreenGrid Scorecard — the workload you'll build + the SaaS prerequisite
11:15	The hack brief, teams & environment check
12:00	Lunch
13:30	Hack · Sprint 1 — clone, run Hello World, render with fake data (steps 1–7)
15:30	Break
15:45	Hack · Sprint 2 — switch to OneLake & polish (steps 8–9)
16:45	Team demos (5 min each)
17:00	Wrap-up, KPIs & next steps

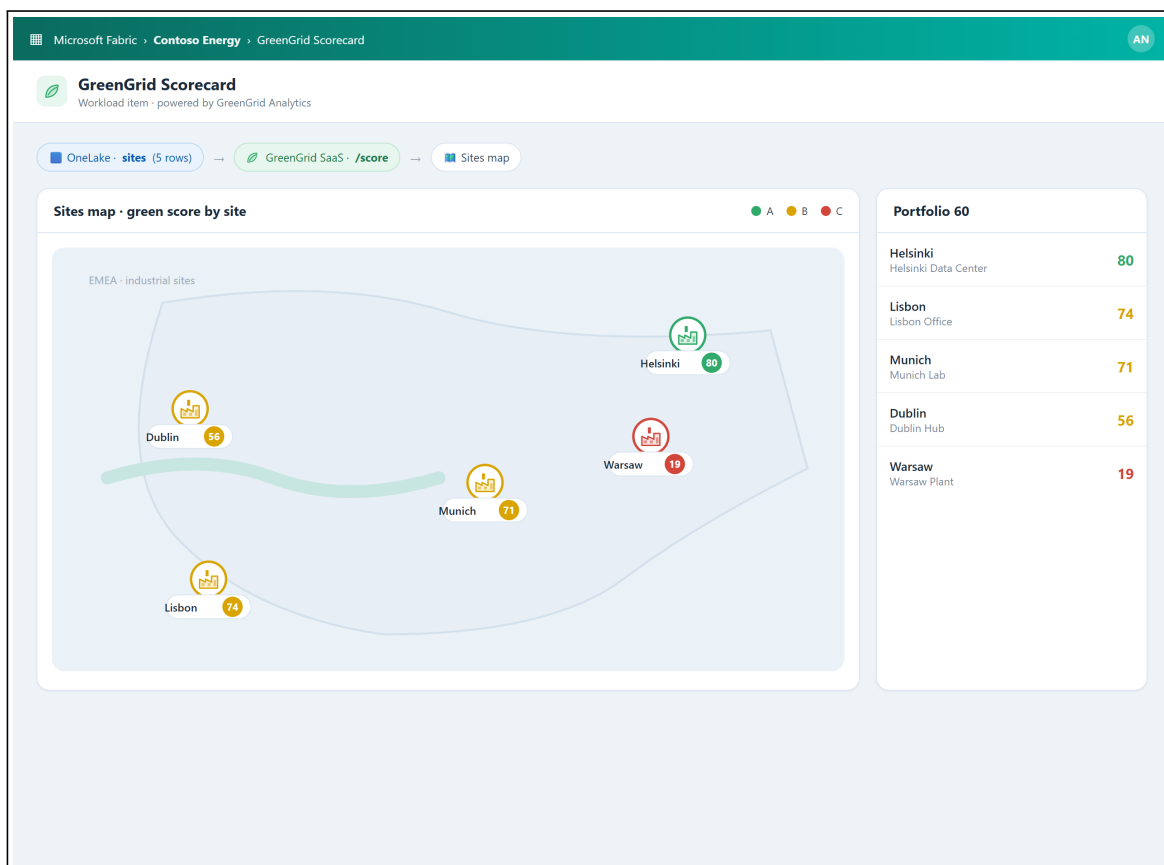
Part A — Understand

1) The scenario in plain words

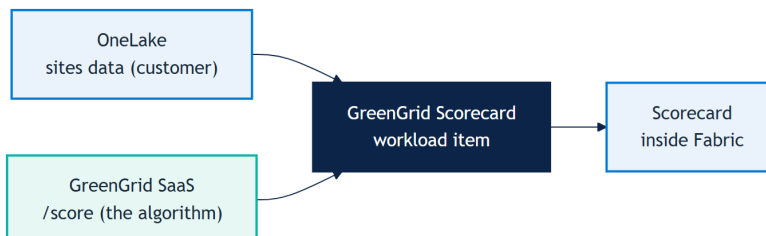
GreenGrid Analytics is a software company (an **SDC** — a solution development company). Its product scores how **sustainable** a customer's sites are, from their energy use and renewable mix. The customer's data already lives in Microsoft Fabric (**OneLake**). Your job: bring GreenGrid's scoring **inside Fabric**, as a native screen, **without moving the data**.

The result looks like this — a sustainability scorecard and a map of the industrial sites, rendered inside Fabric:





2) The three pieces (and who owns the algorithm)



- **OneLake** = where the **customer's data** lives.
- **GreenGrid SaaS** = a small web service that **holds the scoring algorithm** (GreenGrid's IP). *The trainer usually hosts it; if you run solo, you start it yourself in Step 6a — `cd src/workLoadsdsc && npm run saas:start`.*
- **The workload** = the piece **you build today**: a Fabric *item* that reads OneLake, calls the SaaS, and draws the scorecard.

3) Key words (30-second primer)

- **Workload**: a web app that runs *inside* the Fabric portal. You host it, Fabric shows it.
- **Item**: one thing a user creates in a workspace (here: a *GreenGrid Scorecard*).
- **Extensibility Toolkit**: Microsoft's open-source kit + SDK to build such workloads.
- **Dev Gateway**: a local bridge so Fabric can show your workload while it runs on **your machine** — this is what makes a one-day hack possible.
- **OBO token** (On-Behalf-Of): a secure token the workload gets so it can read the user's OneLake data **as that user** — no passwords, no data copies.

4) Prerequisites & environment setup (do this first)

Modeled on Microsoft's [Build LAB514](#) setup. On a lab VM most tools are pre-installed.

Tools you need:

- Node.js 20+ and npm
- **PowerShell 7, .NET (x64), Azure CLI** (the toolkit setup script uses them)
- Visual Studio Code
- **GitHub Copilot CLI** (the copilot terminal agent) — used for the graphical polish step

Sign in (two accounts):

1. **GitHub Copilot** — run `copilot`, then `/login`, choose **GitHub.com**, finish the browser sign-in, confirm “Signed in successfully”, then `/exit`.
2. **Microsoft Fabric** — open `https://app.fabric.microsoft.com` and sign in. (If it shows “Power BI” bottom-left, switch it to “Fabric”.)

From the trainer / your tenant:

- A **Fabric workspace** with a **capacity** assigned (where the workload runs).
- Permission to create an **Entra app** (or one provided) — required even with the Dev Gateway; `Setup.ps1` creates it.
- The **GreenGrid SaaS URL + API key**.

Verify your setup:

```
node --version
npm --version
copilot --version
pwsh --version
```

What you should see. All print a version, you’re signed in to GitHub Copilot and Fabric, and you have the SaaS URL + key in hand.

5) Team framing (fill this in 2 minutes, then build)

- Team name:
- SaaS URL + API key:
- One sentence you’ll say in the demo:

Part B — Build it step by step (full solution)

Golden rule (read this!): first get the toolkit running with the built-in **Hello World** (steps 1–4). Only then build GreenGrid (steps 5–9). And inside GreenGrid, render **fake data first** (step 7), *then* switch to real OneLake data (step 8). Never debug security before something shows on screen.

Every step says **what it is**, **what to type/paste**, and **what you should see**.

Step 1 — Clone the toolkit

What this is. Get Microsoft’s Extensibility Toolkit on your machine. This is a real repository — you clone it and start from it.

```
git clone https://github.com/microsoft/fabric-extensibility-toolkit
cd fabric-extensibility-toolkit
```

What you should see. A `fabric-extensibility-toolkit` folder with `Workload/` and `scripts/` inside.

Step 2 — Configure the dev environment (one script)

What this is. A setup script that creates the Entra app, writes your `.env`, and downloads the Dev Gateway. It asks you to sign in and pick your workspace.

```
cd scripts/Setup
./Setup.ps1 -WorkloadName "Org.GreenGrid"
```

The WorkloadName must look like Organization.Name — use Org.GreenGrid for the hack. On macOS/Linux run `pwsh ./Setup.ps1 -WorkloadName "Org.GreenGrid"`.

Why an Entra app even with the Dev Gateway? The Dev Gateway only **routes** your local workload into Fabric — it does **not** provide identity. Your workload authenticates with **Microsoft Entra**, and the Entra app is what lets it receive Fabric tokens and acquire the **OBO token** to read OneLake (step 8). Setup.ps1 creates this app for you (or reuses an existing one).

What you should see. The script ends with “setup complete” and prints the next steps.

Step 3 — Start the dev server + the Dev Gateway

What this is. Two long-running processes, in **two terminals**. One serves your workload UI from localhost; the other connects Fabric to that localhost.

```
# Terminal 1 – the frontend + APIs
cd scripts/Run
./StartDevServer.ps1
```

```
# Terminal 2 – the bridge to Fabric
cd scripts/Run
./StartDevGateway.ps1
```

Then, in the **Fabric portal**: open **Settings -> Admin portal** and enable the developer tenant settings, then **Settings -> Developer settings -> enable Fabric Developer Mode**.

What you should see. Both terminals keep running without errors, and Developer Mode is ON in Fabric.

If StartDevGateway.ps1 fails with Dev instance registration ... Forbidden, error-Code: FeatureNotAvailable : your tenant hasn't enabled customer-developed workloads. An **admin** must turn on, under **Admin portal -> Tenant settings -> Additional workloads: Capacity admins and contributors can add and remove additional workloads, Workspace admins can develop workloads, and Users can see and work with additional workloads not validated by Microsoft** — **and** the workspace must be on a **Fabric/Trial capacity** (not Pro/PPU). Full steps in the setup guide (§6.1). It is not a code issue — the gateway is refused by Fabric.

Not a tenant admin? You can't self-enable it — either ask your admin for those toggles, run the hack in a tenant **you** administer (a Fabric Trial / demo tenant), or keep building locally without Fabric: `npm run saas:start + npm run demo:build`, then open `src/workloadsdc/demo/out/scorecard.html` (same algorithm, no Fabric needed). See setup §6.1.1.

Step 4 — Hello World test (prove the gateway works, before building anything)

What this is. The toolkit ships a **Hello World** item. Create it in Fabric to confirm the whole chain — *your localhost code -> Dev Gateway -> Fabric* — works. **This is your safety checkpoint.**

1. Go to <https://app.fabric.microsoft.com/workloadhub/detail/Org.GreenGrid.Product?experience=developer> (the workload shows as “Hello Fabric!” until you rename it).
2. Click the **Hello World** item type on the left.
3. Pick your **dev workspace** in the dialog.
4. The Hello World editor opens.

What you should see. The Hello World editor renders **inside Fabric**, served from your machine. If you see it, everything works — now you can build GreenGrid. (*Nothing shows up? Developer Mode is off, or one of the two terminals isn't running.*)

Step 5 — Create the GreenGrid item

What this is. Generate a fresh item to hold your scorecard (don't overwrite Hello World).

```
cd scripts/Setup
./CreateNewItem.ps1 -Name "GreenGridScorecard"
```

This creates `Workload/app/items/GreenGridScorecardItem/` on your machine, including **GreenGridScorecardItemEditor.tsx** — the **React component Fabric renders when your item is opened** (this is “the editor”; it starts nearly empty).

CreateNewItem.ps1 scaffolds the code but leaves 4 things for you to wire up — do all four now, or the Dev Gateway rejects the item / Fabric won't surface it.

1. Register it in ITEM_NAMES (else it isn't packaged -> not in the manifest). Edit **all three** `Workload/.env.dev`, `.env.test`, `.env.prod` and append the `-Name` value:

```
- ITEM_NAMES=HelloWorld
+ ITEM_NAMES=HelloWorld,GreenGridScorecard
```

2. Add the locale keys (else Missing keys in en-US locale: `..._DisplayName`, `..._DisplayName_Plural`). In `Workload/Manifest/assets/locales/en-US/translations.json` add three keys (`Display/Plural` are used by the item manifest, `Description` by the create-card in step 4):

```
"GreenGridScorecardItem_DisplayName": "GreenGrid Scorecard",
"GreenGridScorecardItem_DisplayName_Plural": "GreenGrid Scorecards",
"GreenGridScorecardItem_Description": "Score the sustainability of your
  ↩ sites in Fabric."
```

Every key must be **referenced somewhere** — an unreferenced key gives `Unused keys in en-US locale`.

3. Add the item icon (else `MissingAssets: 'assets/images/GreenGridScorecardItem_Icon.png'`). Drop a 48×48 PNG named `<Item>_Icon.png` into `Workload/Manifest/assets/images/` (quickest fix: copy `HelloWorldItem_Icon.png`, swap a nicer icon later).

4. Add a create-card to Product.json (else the gateway registers fine and Hello World shows, but **your item never appears in New item**). In `Workload/Manifest/Product.json` add your item to `homePage.recommendedItemTypes` **and** add a card to `createExperience.cards` mirroring the `HelloWorld` one — set `title/description/icon` to your `GreenGridScorecardItem_*` keys/asset, `availableIn`: `["home", "create-hub", "workspace-plus-new", "workspace-plus-new-teams"]`, and `"itemType": "GreenGridScorecard"`.

Restart both terminals so the manifest is rebuilt and re-delivered to Fabric — the Dev Server (Ctrl+C, `./StartDevServer.ps1`) **and** the Dev Gateway (Ctrl+C, `./StartDevGateway.ps1`). On restart the gateway should log `Configured items from ITEM_NAMES: HelloWorldItem, GreenGridScorecardItem` and `Registering dev instance...` should **succeed** (no fail).

Now **create an instance** of your item *in the Fabric portal* so you have something to render into:

1. **Refresh** Fabric (F5), then open the Workload Hub: <https://app.fabric.microsoft.com/workloadhub/detail/developer> (same place as Hello World in Step 4).
2. Click the **GreenGridScorecard** item type -> pick your **dev workspace** -> the editor opens.

Type vs instance. `CreateNewItem.ps1` scaffolds the item **type + its editor code** (on your machine). Creating it in Fabric makes an **instance** that runs that code inside Fabric. Keep this instance open: every time you save the editor `.tsx`, the dev server hot-reloads and this open item re-renders — you do **not** re-create the item for each change.

Don't see GreenGridScorecard in “New item”? 99% of the time it's the `ITEM_NAMES` step above: confirm the item is added in **all three** `.env` files, restart **both** the Dev Server **and** the Dev Gateway, then **refresh** Fabric. (*The `BuildManifestPackage ... CursorPosition: The handle is invalid` warning is harmless — it's a console-cursor quirk; `Delivered manifest package successfully` means the build is fine.*)

What you should see. An empty `GreenGridScorecard` editor opens in Fabric (your code, rendered in the portal).

Step 6 — Start & check the GreenGrid SaaS

What this is. The SaaS holds GreenGrid's scoring algorithm — your workload is useless without it, so it must be **running and reachable** before you build.

6a. Start the SaaS

If the **trainer hosts it**, just grab the **SaaS URL + API key** they give you and skip to 6b.

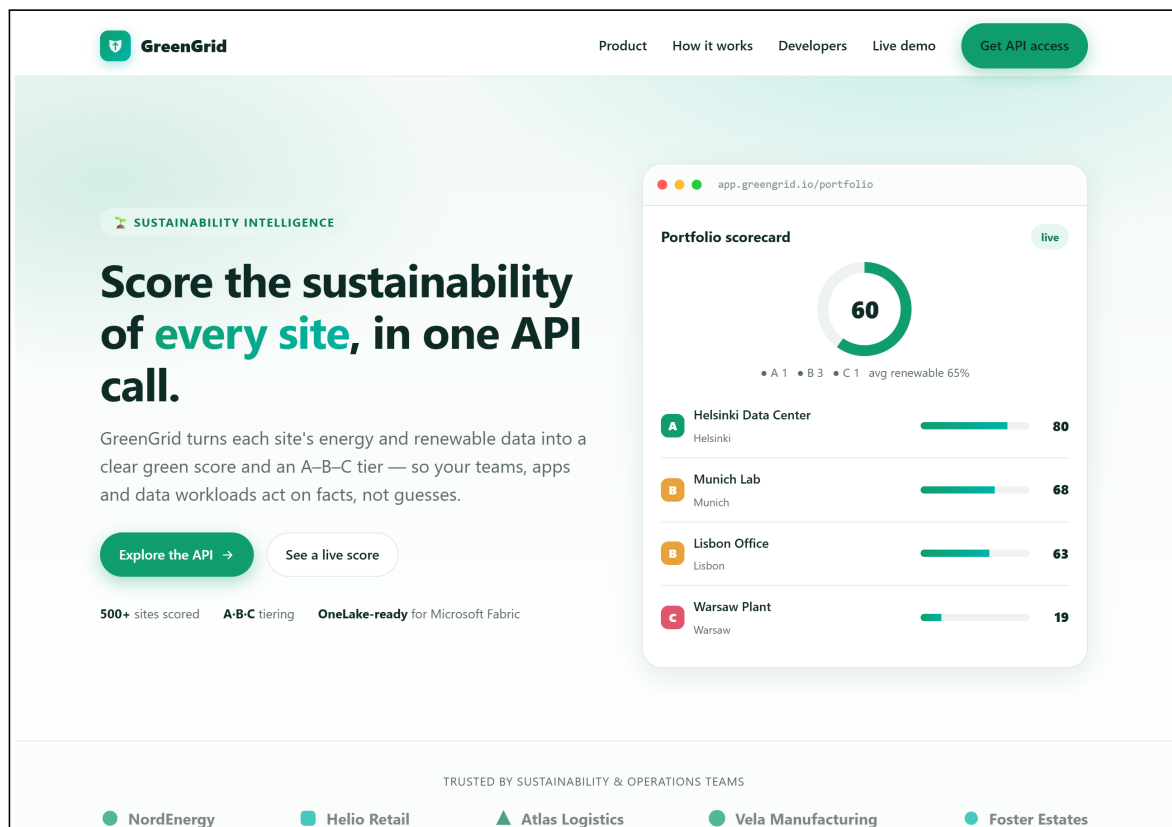
If you're **running it yourself** (solo, or as a local fallback), start it from the kit:

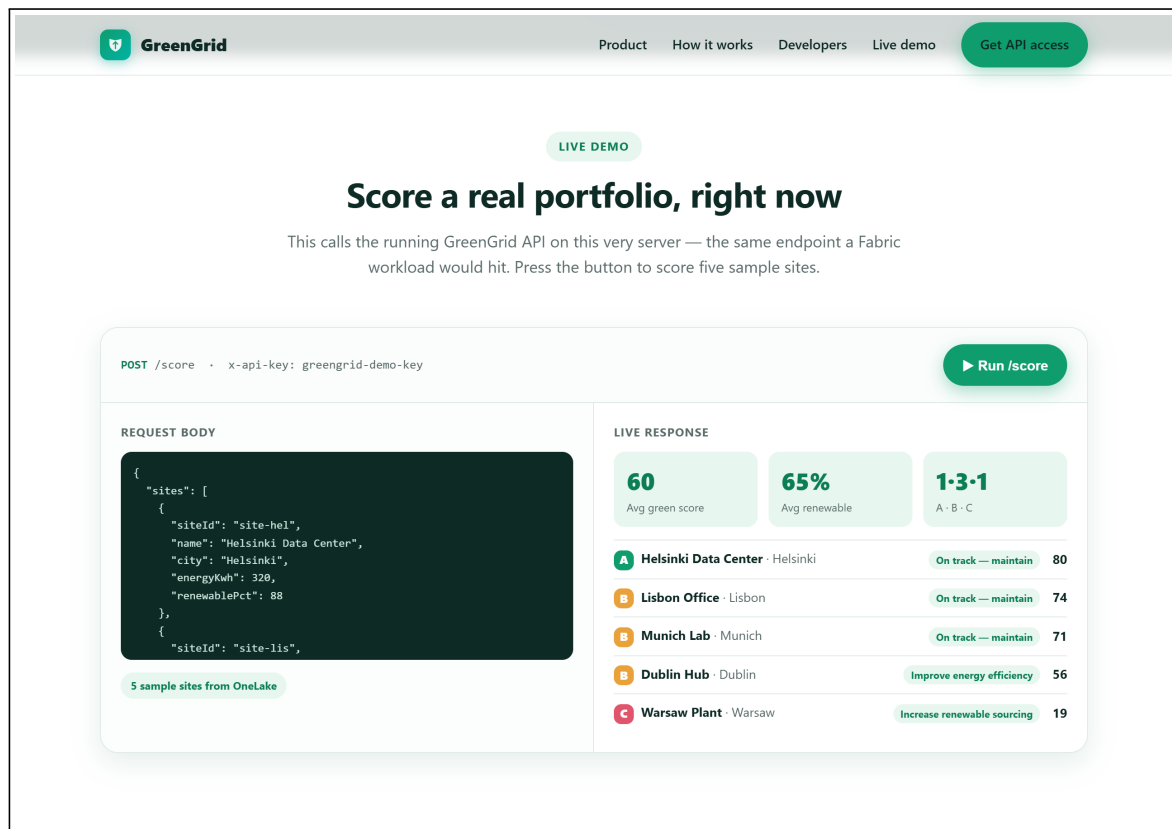
```
cd src/workloadsdc
npm install
npm run saas:start
```

It starts on **http://localhost:8787** with the default API key **greengrid-demo-key** (override with the `GREENGRID_API_KEY` env var). The console should print:

```
GreenGrid SaaS listening on http://localhost:8787
GET / (marketing site + live API demo)
GET /health
POST /score (header: x-api-key: greengrid-demo-key)
```

Keep this terminal open — the SaaS must stay running for the whole hack. Open **http://localhost:8787/** in a browser to see the GreenGrid website + a **live /score demo** (same algorithm your workload will call).





6b. Check it's alive

Replace `<SAAS_URL>` with your URL (local: `http://localhost:8787`) and `<API_KEY>` with your key (local: `greengrid-demo-key`):

```
curl -i <SAAS_URL>/health # expect: 200
curl -s -X POST <SAAS_URL>/score \
  -H "Content-Type: application/json" -H "x-api-key: <API_KEY>" \
  -d '{"sites":[{"siteId":"s1","name":"Helsinki
    ↪ DC","city":"Helsinki","energyKwh":320,"renewablePct":88}]}'
```

On Windows PowerShell, `curl` is an alias for `Invoke-WebRequest` and won't accept the flags above — use `curl.exe` (real curl), or the native form:

```
Invoke-RestMethod http://localhost:8787/health
Invoke-RestMethod -Method Post -Uri http://localhost:8787/score `
  -Headers @{ 'x-api-key' = 'greengrid-demo-key' } -ContentType
    ↪ 'application/json' `
  -Body '{"sites":[{"siteId":"s1","name":"Helsinki
    ↪ DC","city":"Helsinki","energyKwh":320,"renewablePct":88}]}'
```

What you should see. `/health` returns 200, and `/score` returns a greenScore and a tier (Helsinki -> greenScore 80, tier A). Note your working **SAAS_URL + API_KEY** — you'll paste them into the workload client in Step 7.

Step 7 — Make it work with fake data (Milestone M1)

Create **four files** in `Workload/app/items/GreenGridScorecardItem/`, then render the scorecard from the generated editor. This proves the whole chain *screen -> SaaS -> screen*.

7a. contracts.ts — the data shapes

```
export type SiteRecord = {
  siteId: string; name: string; city: string; energyKwh: number; renewablePct:
  ↪ number;
};
export type GreenTier = 'A' | 'B' | 'C';
export type ScoredSite = SiteRecord & {
  efficiency: number; greenScore: number; tier: GreenTier; tip: string;
};
export type ScoreResponse = {
  sites: ScoredSite[];
  summary: { totalSites: number; avgScore: number; avgRenewablePct: number;
    tierCounts: Record<GreenTier, number>; best: ScoredSite; worst: ScoredSite; };
};
```

7b. greengridClient.ts — call the SaaS

```
import type { ScoreResponse, SiteRecord } from './contracts';

const SAAS_URL = 'http://localhost:8787'; // <- replace with the trainer URL
const API_KEY = 'greengrid-demo-key'; // <- replace with the trainer key

export async function scorePortfolio(sites: SiteRecord[]): Promise<ScoreResponse> {
  const res = await fetch(`${SAAS_URL}/score`, {
    method: 'POST',
    headers: { 'Content-Type': 'application/json', 'x-api-key': API_KEY },
    body: JSON.stringify({ sites }),
  });
  if (!res.ok) throw new Error(`GreenGrid SaaS error ${res.status}`);
  return res.json();
}
```

7c. seed.ts — fake sites to start

```
import type { SiteRecord } from './contracts';
export const seedSites: SiteRecord[] = [
  { siteId: 'site-hel', name: 'Helsinki Data Center', city: 'Helsinki', energyKwh:
  ↪ 320, renewablePct: 88 },
  { siteId: 'site-lis', name: 'Lisbon Office', city: 'Lisbon', energyKwh: 210,
  ↪ renewablePct: 71 },
  { siteId: 'site-muc', name: 'Munich Lab', city: 'Munich', energyKwh: 430,
  ↪ renewablePct: 80 },
  { siteId: 'site-dub', name: 'Dublin Hub', city: 'Dublin', energyKwh: 540,
  ↪ renewablePct: 62 },
  { siteId: 'site-waw', name: 'Warsaw Plant', city: 'Warsaw', energyKwh: 880,
  ↪ renewablePct: 24 },
];
```

7d. Scorecard.tsx — the screen

It takes a `getSites` function, so swapping the data source later is **one line**.

```
import React, { useEffect, useState } from 'react';
import { FluentProvider, webLightTheme, Spinner, Badge } from
  ↪ '@fluentui/react-components';
import { scorePortfolio } from './greengridClient';
import type { ScoreResponse, SiteRecord } from './contracts';
```

```

const tierColor = (t: 'A' | 'B' | 'C') => (t === 'A' ? '#2faa6a' : t === 'B' ?
  ↪ '#d9a300' : '#d1453b');

export function Scorecard({ getSites }: { getSites: () => Promise<SiteRecord[]> }) {
  const [data, setData] = useState<ScoreResponse | null>(null);
  const [error, setError] = useState<string | null>(null);

  useEffect(() => {
    getSites().then(scorePortfolio).then(setData).catch((e) =>
  ↪ setError(String(e)));
  }, [getSites]);

  if (error) return <div role="alert">Failed: {error}</div>;
  if (!data) return <Spinner label="Scoring sites..." />;

  return (
    <FluentProvider theme={webLightTheme}>
      <div style={{ padding: 24 }}>
        <h1>GreenGrid Scorecard</h1>
        <p>Data from OneLake, scored by GreenGrid</p>
        <div style={{ fontSize: 48, fontWeight: 700 }}>{data.summary.avgScore}<span
  ↪ style={{ fontSize: 18 }}> / 100</span></div>
        <div style={{ display: 'grid', gridTemplateColumns: 'repeat(2,1fr)', gap:
  ↪ 12, marginTop: 16 }}>
          {data.sites.map((s) => (
            <div key={s.siteId} style={{ border: '1px solid #dbe6f1', borderRadius:
  ↪ 12, padding: 14 }}>
              <div style={{ display: 'flex', justifyContent: 'space-between' }}>
                <strong>{s.name}</strong>
                <Badge appearance="filled" style={{ background: tierColor(s.tier)
  ↪ }}>Tier {s.tier}</Badge>
              </div>
                <div style={{ color: '#7a8a9a', fontSize: 12 }}>{s.city}</div>
                <div style={{ fontSize: 26, fontWeight: 700 }}>{s.greenScore}</div>
                <div style={{ fontSize: 13 }}>{s.tip}</div>
              </div>
            </div>
          ))}
        </div>
      </FluentProvider>
    );
}

```

7e. Render it from the generated editor

GreenGridScorecardItemEditor.tsx is the file `CreateNewItem.ps1` generated in **Step 5** — the component Fabric shows when your item opens. Open it and render your Scorecard with the **fake** seed data (replace the placeholder the toolkit put in the editor's JSX):

```

import { Scorecard } from './Scorecard';
import { seedSites } from './seed';

// inside the editor component's returned JSX:
<Scorecard getSites={() => Promise.resolve(seedSites)} />

```

Save the file. The dev server hot-reloads, so the GreenGridScorecard **item you already opened in Fabric** (Step 5) now shows the scorecard — no need to re-create it. *(If you closed it, reopen it from the Workload Hub as in Step 5.)*

What you should see — Milestone M1. The item shows the portfolio score and 5 site cards with their tier, **inside Fabric**. The whole chain works on fake data. This is demoable.

Step 8 — Switch to real OneLake data (Milestone M2)

What this is. Replace the fake seedSites with the customer's **real** site data in OneLake.

Where does the data live?

In Fabric, customer data lives in a **Lakehouse** inside OneLake. A Lakehouse has **Files** (raw files: CSV, Parquet...) and **Tables** (Delta). For this hack the sites are a **CSV file**: Files/sites.csv. *(A CSV is the simplest thing to read from a workload; a Delta table would need a SQL endpoint — out of scope for one day.)*

Format of Files/sites.csv (a ready copy is in the kit at src/workloadsdc/data/sites.csv):

```
siteId,name,city,energyKwh,renewablePct
site-hel,Helsinki Data Center,Helsinki,320,88
site-lis,Lisbon Office,Lisbon,210,71
site-muc,Munich Lab,Munich,430,80
site-dub,Dublin Hub,Dublin,540,62
site-waw,Warsaw Plant,Warsaw,880,24
```

The trainer pre-loads this file (Lakehouse -> **Files** -> **Upload**). To test your own, upload a CSV with the **same header** to Files/.

You also need two IDs (visible in the Lakehouse URL, or from the item context): workspaceId and lakehouseId.

8a. onelake.ts — read & parse the CSV

```
import type { WorkloadClientAPI } from '@ms-fabric/workload-client';
import type { SiteRecord } from './contracts';

export async function readSites(
  client: WorkloadClientAPI, workspaceId: string, lakehouseId: string
): Promise<SiteRecord[]> {
  // 1) Token to read OneLake AS the signed-in user (OBO).
  const { token } = await client.auth.acquireAccessToken({
    additionalScopesToConsent: ['https://storage.azure.com/user_impersonation'],
  });
  // 2) Read the CSV file from the Lakehouse "Files" area.
  const url = `https://onelake.dfs.fabric.microsoft.com/${workspaceId}/${lakehouseId}/Files/sites.csv`;
  const res = await fetch(url, { headers: { Authorization: `Bearer ${token}` } });
  if (!res.ok) throw new Error(`OneLake read failed: ${res.status}`);
  // 3) Parse CSV text into rows.
  const [header, ...lines] = (await res.text()).trim().split(/\r?\n/);
  const cols = header.split(',').map((c) => c.trim());
  return lines.filter(Boolean).map((line) => {
    const cells = line.split(',');
    const row = Object.fromEntries(cols.map((c, i) => [c, (cells[i] ?? '').trim()]));
    return { siteId: row.siteId, name: row.name, city: row.city,
      energyKwh: Number(row.energyKwh), renewablePct: Number(row.renewablePct) };
  });
}
```

8b. Swap one line in the editor

The generated editor exposes the workload client and the item's context. Change the render to:

```
import { readSites } from './onelake';

// before (fake):
```

```
<Scorecard getSites={() => Promise.resolve(seedSites)} />
```

```
// after (real OneLake CSV):
```

```
<Scorecard getSites={() => readSites(workloadClient, workspaceId, lakehouseId)} />
```

What you should see — Milestone M2. The same scorecard, now built from the **OneLake CSV**. (If the OBO token fights you, keep the fake-data line as a fallback and keep moving.)

Step 9 — Make it graphical (Milestone M3)

What this is. Upgrade the plain list into the polished scorecard + map. Easiest with GitHub Copilot (the toolkit ships AI instructions in `.github/copilot-instructions.md`):

Upgrade the GreenGrid Scorecard screen: add a circular green-score gauge for the

↪ portfolio,

an A/B/C tier distribution, provenance chips "Data from OneLake" then "Scored by

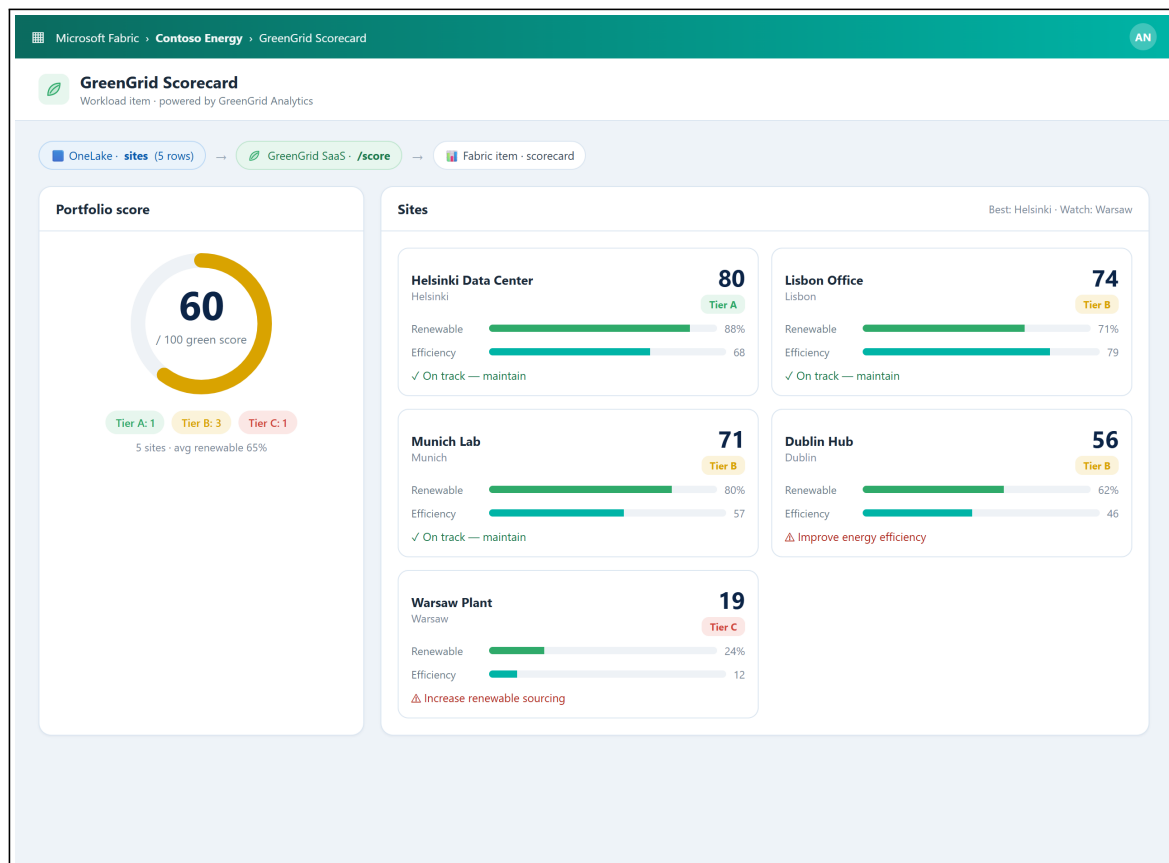
↪ GreenGrid",

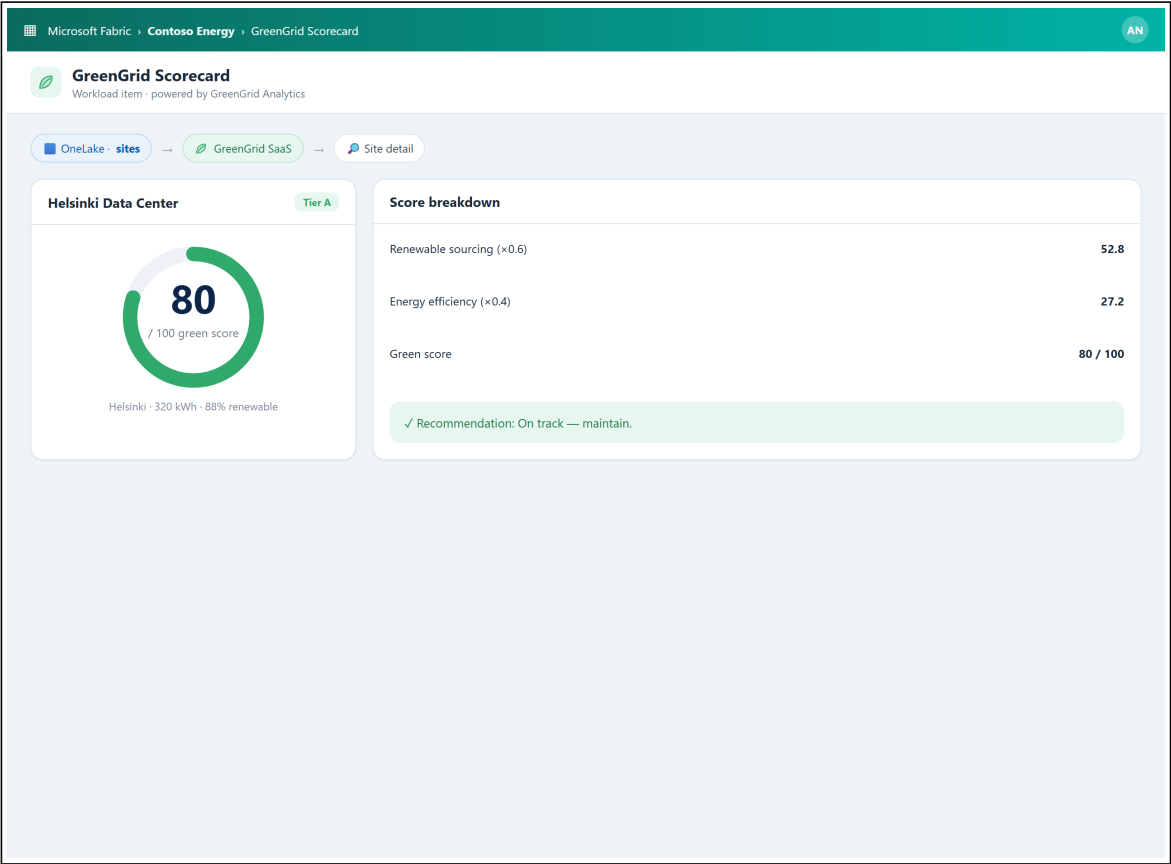
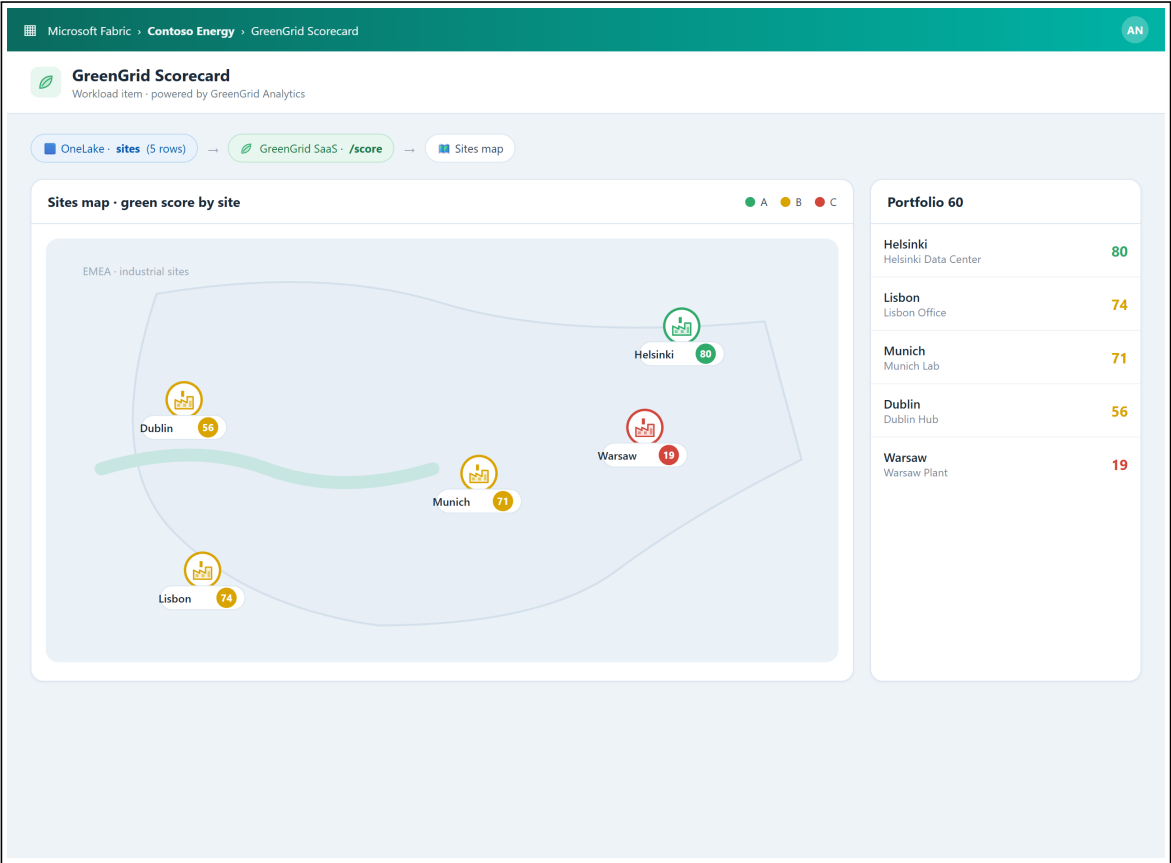
and a second view showing the sites as factory markers on a simple map, colored by

↪ tier.

Style: Fluent UI, teal #00b4a6 + green accents. Keep the `getSites` data flow.

What you should see — Milestone M3. A gauge, A/B/C tiers and the sites map, rendered natively inside Fabric:





Wrap-up

Demo storyboard (5 minutes)

1. **Context** (30s): the SaaS value (the algorithm) you brought into Fabric.
2. **Flow** (3m): open the item, read OneLake, call the SaaS, show the scorecard + map.
3. **Architecture** (45s): OneLake (data) + SaaS (algorithm) + workload (native UI).
4. **Publish plan** (45s): from Dev Gateway to a released workload.

Reflection

- Where did the OneLake -> SaaS contract surprise you?
- What stays in the SaaS (the IP) vs. in the workload (the experience)?
- What would you harden before production?

Reference assets

- Full solution code: `src/workloadsgdc/`
- SaaS prerequisite: `src/workloadsgdc/saas/`
- Sample OneLake data: `src/workloadsgdc/data/sites.csv`
- Trainer answer key + screenshots: `docs/microhack-2-gdc-workloads-solution.md`